
SIMPLE ALGORITHM OF SLIDING AVERAGES WEB APP

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ABSTRACT

I present the web application of a simple sampling algorithm (SALSA) originally conceived 24 years ago, and published in 2007. New additional features are incorporated to enable a user to better perform experiments using the algorithm. Some additional insights and discussion on the design and implementation of the algorithm itself are provided. Results of further experiments using the application are to be presented in separate follow-up manuscripts.

Keywords sequence analysis · bioinformatics

1 Introduction

SALSA was designed to run a sliding window across the amino acid sequence of natively unfolded proteins involved in human neurodegenerative diseases. taking the mean of an unspecified property over each window. There are three aspects of this algorithm that make it unique, particularly in their combination. Firstly, although the user can select to use just one window size, the algorithm is intended to be run on one sequence using a range of window sizes. This generates a much larger number of windows and their scores. The other two aspects concern the manner in which the window scores are processed: the mean score of a window is assigned to every member of that window. In the example of a protein sequence, if the mean score of a window at positions 1 to 5 (MDVFM) is 1.3, each residue in this window is assigned 1.3. Hence M1, D2, V3, F4, and M5 each hold the score 1.3 in this specific window. This is in contrast to, for example, assigning the score only to the central position. The final step involves taking the sum of all of the scores for each

2 Headings: first level

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2.1 Headings: second level

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*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

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$$\xi_{ij}(t) = P(x_t = i, x_{t+1} = j | y, v, w; \theta) = \frac{\alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})}{\sum_{i=1}^N \sum_{j=1}^N \alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})} \quad (1)$$

2.1.1 Headings: third level

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3 Examples of citations, figures, tables, references

3.1 Citations

Citations use natbib. The documentation may be found at

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf>

Here is an example usage of the two main commands (`citet` and `citep`): Some people thought a thing [Kour and Saabne, 2014a, Hadash et al., 2018] but other people thought something else [Kour and Saabne, 2014b]. Many people have speculated that if we knew exactly why Kour and Saabne [2014b] thought this...

3.2 Figures

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3.3 Tables

See awesome Table 1.

The documentation for booktabs (‘Publication quality tables in LaTeX’) is available from:

<https://www.ctan.org/pkg/booktabs>

3.4 Lists

- Lorem ipsum dolor sit amet
- consectetur adipiscing elit.
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²Sample of the first footnote.

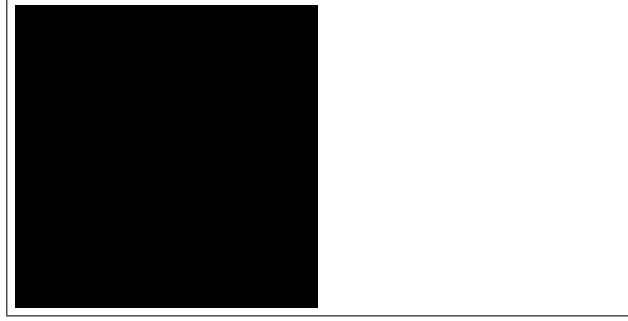


Figure 1: Sample figure caption.

Table 1: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

References

- George Kour and Raid Saabne. Real-time segmentation of on-line handwritten arabic script. In *Frontiers in Handwriting Recognition (ICFHR), 2014 14th International Conference on*, pages 417–422. IEEE, 2014a.
- Guy Hadash, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.
- George Kour and Raid Saabne. Fast classification of handwritten on-line arabic characters. In *Soft Computing and Pattern Recognition (SoCPaR), 2014 6th International Conference of*, pages 312–318. IEEE, 2014b. doi:10.1109/SOCPAR.2014.7008025.